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Calling Behavior and Localization of Blue Whales in Southern California

A Thesis submitted in partial satisfaction of the requirements
for the degree Master of Science

in

Marine Biology

by

Gabrielle Marie Arrieta

Committee in charge:

Simone Baumann-Pickering, Chair
Kaitlin Frasier
John Hildebrand

2023

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The Thesis of Gabrielle Marie Arrieta is approved, and it is acceptable in quality and form for publication on microfilm and electronically.

University of California San Diego

2023

DEDICATION

This master's thesis is dedicated to my mother. For every 2 hour drive she took me on to Monterey Bay Aquarium so I could volunteer and every time she tried to convince me to become a lawyer. Mumsies, you are the reason I am who I am and why I was able to get here. Love you to the moon and back.

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LIST OF ABBREVIATIONS

TDOA	Time difference of arrival
HARP	High frequency acoustic recording package
HE	Site H East
HW	Site H West
HS	Site H South
HN	Site H North
Pk-pk/pp	Peak to Peak
0-pk	Zero to peak
Rms	Root mean square
CTD	Conductivity, temperature and depth
SOCAL	Southern California

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I would also like to thank several graduate students and colleagues who were always willing to offer me advice. Lauren Baggett and Clara Schoenbeck for being the best office mates ever and taking on this masters journey with me. Shannon Dolan and Jack Ewing for their constant advice and support. Ella Kim, Michaela Alksne, Eric Snyder and many more for their endless wisdom and advice on studying acoustics.

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ABSTRACT OF THE THESIS

Calling Behavior and Localization of Blue Whales in Southern California

by

Gabrielle Marie Arrieta

Master of Science in Marine Biology

University of California San Diego, 2023

Simone Baumann-Pickering, Chair

Blue whales, *Balaenoptera musculus*, are the largest animals on Earth and yet there is much unknown about their life history. Blue whales produce low frequency sounds including short (1-4 s) down-sweeping D calls, commonly associated with foraging. This study used long term passive acoustic monitoring to investigate a new sequence of D calls observed in data

collected from 2007 to 2020 in Southern California and localized sequences on July 22nd 2019. On that day, 28 sequences occurred in a two hour and twenty-minute period. The D calls during this time had an average 7.3 s +/- 1.7 s inter-call interval within a sequence and 7 calls +/- 1 call per sequence. In the data from 2007 to 2020, sequences were present in all the years with a peak in sequences per day in 2019 with the maximum number of sequences per year also occurring in 2019. To understand how sequences were produced spatially, the D calls on July 22nd 2019 were localized. This resulted in 194 locations for all D calls and an average location for each sequence. Average locations were scattered within a 250 m by 350 m area, indicating that an individual whale may be producing the sequences. Using the location for each call, source level was computed using received level and transmission loss. The average root mean square source level calculated over 30 to 80 Hz was 162.3 +/- 2.2 dB re 1µPa at 1 m. By combining both observations of calling behavior and localizations, we can gather insight into the temporal and spatial behavior of blue whales when emitting D calls.

INTRODUCTION

North Pacific blue whales can be found off the coast of Southern California during their annual migration from South America to the Gulf of Alaska (Hazen et al. 2017). Much of what we know about their movement during this time is from visual sightings or tags attached to the whales (Mate et. al 1999; Calambokidis et. al 2007; Hazen et. al 2017; Palacios et. al 2019). Since blue whales produce sound, we can also use autonomous passive acoustics to study these animals as a non-invasive alternative. Previous research has described three different blue whale sounds in the Eastern North Pacific (Thompson et al. 1996; Oleson et al. 2007a; McDonald et al 2001). A and B calls have multiple harmonics usually below 150 Hz (McDonald et al 2001). These calls can be 10 to 20 s long and are produced either in a sequence to form song or singularly. The third call type is a D call which is a down sweeping call that occurs in the 30 to 120 Hz frequency band and can be between 1 and 4 s long (Thompson et al 1996; Oleson et al. 2007a). D calls are highly variable in frequency content and sweep slope (Oleson et al 2007b). D calls have been produced during shallow dives (less than 35 m) while engaging in deeper, presumable foraging dives in between D calling. D calls are produced by both sexes, usually when there is more than one individual present, indicating a link to social interaction rather than reproduction (Oleson et al 2007b). Unlike A and B calls, D calls have not been shown to be produced in regular sequences to form song. By studying blue whale D calls and their distinctive characteristics, we aimed to understand the behavioral context of these calls.

Passive acoustics can be used to determine the location of a calling whale when there are multiple recording instruments in the water (Wiggins et al. 2013; Wiggins and Hildebrand 2020; Varga et al. 2018; Soldevilla et al. 2022; Miller et al. 2021). Localization allows us to calculate the location of a whale for each call to observe the movement of the whale. For low frequency

calls, large aperture arrays are used to localize calls using a sequence of steps. First, the call is detected on one of the instruments so that the same call can be compared with multiple instruments. Detections are done either using a manual or automated method depending on the call type and the automatic detectors available at the time of the study. Other requirements for blue whale localization include accurate receiver locations, source depth, synchronized clocks, and an accurate sound speed. Once all these requirements are satisfied, time difference of arrivals can be computed and then locations of the whale producing each call.

Blue whales have been acoustically localized before using a variety of methods. Blue whale B calls have been used to localize blue whales in the Northeast Pacific using the Navy's Sound Surveillance System (SOSUS). This study resulted in locations that were consistent with visual sightings and confirmed the ability of using passive acoustics to determine location (Stafford et al. 1998). Using hyperbolic localization and time difference of arrivals, blue whale B calls were localized in the Southern Ocean with bottom-moored hydrophones (Širović et al. 2007). Thes studies and many more have used blue whale song to localize these whales in different parts of the world, however D calls have not been localized extensively. A study in the Antarctic used directional sonobuoys to estimate the locations of whales and determine D call source levels. D calls were localized if they were detected on multiple sonobuoys using time difference of arrivals, which resulted in clusters of locations on different days (Miller et al 2021). Despite the unique shape of D calls that make it easy to detect the same call on multiple instruments, D calls have never been used to localize blue whales using fixed hydrophones. Locations allow us to compute source levels for individual blue whales which are important for density estimation analyses, although we do not yet have a reliable source level estimate for the North Pacific blue whale populations.

Blue whales are protected under the Endangered Species Act and Marine Mammal Protection Act after their population was greatly affected by whaling (NOAA Fisheries, 2022). This study focuses on the Eastern North Pacific stock of blue whales which according to the most recent stock assessment show an increase in population size since the 1990s (NOAA Fisheries, 2022). The lower 20th percentile value stock estimate from mark and recapture methods is 1,767 whales (Calambokidis and Barlow, 2020). Although other populations of blue whales do not show as much recovery (Branch et al. 2004; Williams et al. 2011), The Eastern North Pacific population seems to have recovered from whaling (Monnahan et al. 2015). The continual monitoring of this population is important as blue whales serve as indicators of the health of the ecosystem (Wachtendonk et al 2022). By analyzing call behavior and determining the location of different call types, it is possible to understand the behavior of blue whales in Southern California and their response to human activities. With the threat of climate change, it is important to get a baseline understanding of the movement of these large creatures and how this ecosystem is important to them. In this study, acoustic data from 2007 to 2020 is used to analyze D call sequences and localize the sequences to understand their spatial distribution.

METHODS

i. Site Information

Site H (32° 51.262' N 119° 08.938' W) was located west of San Clemente Island and on the western side of the San Nicolas Basin (Figure 1a). In this analysis we use acoustic data from July 24th 2007 to October 29th 2020 at this site recorded on HARPs or High Frequency Acoustic Recording Packages (Wiggins and Hildebrand 2007). These HARPs consist of a hydrophone, floats, a datalogger, acoustic releases and weights that keep them moored on the seafloor. All acoustic data was decimated by a factor of 100 from 200 kHz to a sampling rate of 2000 Hz. From July 13th 2019, to July 28th 2019 there were 4 HARPs recording at site H: HE (32° 51.694 N, 119° 8.361 W), HW (32° 51.236 N, 119° 8.930 W), HS (32° 51.130 N, 119° 8.294 W), and HN (32° 51.798 N, 119° 8.813 W) (Figure 1b).

All sites were approximately 1000 m away from each other with the shortest distance between HE and HN at 730 m (Figure 1b). For localization, the instrument positions need to be as accurate as possible, so all locations used were localized using a transponder and transceiver. Upon deployment of each mooring, the ship was maneuvered around the drop site in a circle with a radius of approximately 1200 m. Using travel time between the transponder located on the mooring and the transceiver located on the ship, it was possible to calculate a slant range. Using many slant ranges and GPS positions while circumnavigating the drop location, we were able to calculate the precise landing location of each of the moorings. From this we were able to infer the depth of the hydrophones which were used in the localization (Table 1).

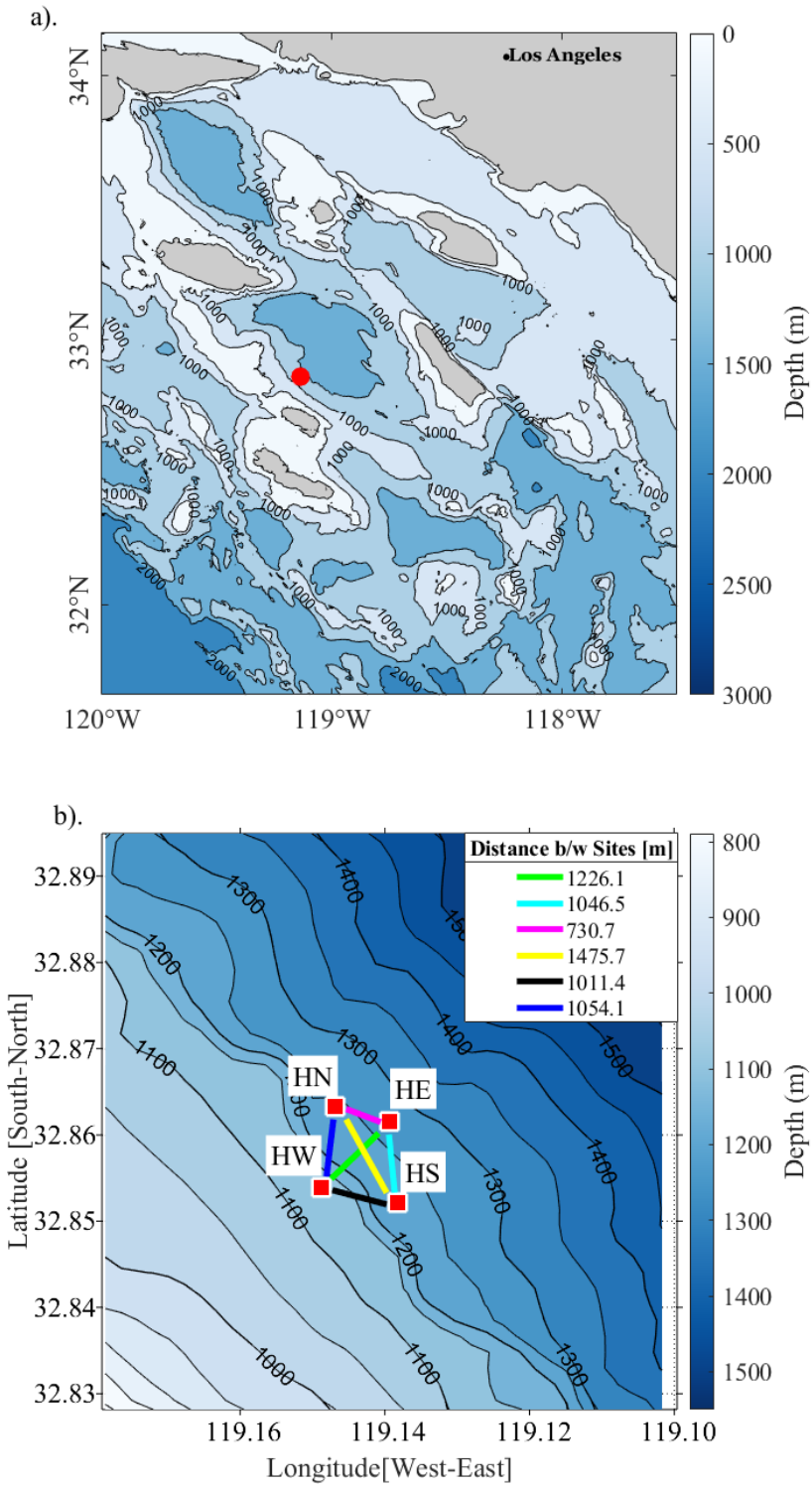

Figure 1: (a) Map of site H. Site H (red dot) location in relation to the Southern California coastline. **(b) Individual instrument locations.** Sites in July 2019 (red boxes) at site H. Color indicates depth and bathymetry contours are every 50 m. Distances between sites are shown with different colored lines connecting two sites.

Table 1: Acoustic data available and location for large aperture array. Each site was named based on its relative location in the array (E = East, W = West, N = North and S = South). Data start and end times are the times in which there was acoustic data available at each site location.

Site	Data Start	Data End	Latitude (N)	Longitude (W)	Hydrophone Depth (m)
HE	06/01/2019	12/10/2019	32° 51.694	-119° 8.361	1253
HW	07/14/2019	11/9/2019	32° 51.236	-119° 8.930	1113
HS	05/05/2019	07/28/2019	32° 51.130	-119° 8.294	1234
HN	05/05/2019	07/28/2019	32° 51.798	-119° 8.813	1243

ii. Detection of calls

D calls were detected from 2007 to 2020 using a mix of automated detection and manual evaluation. A generalized power-law detection algorithm created by Helble et al (2012) to detect humpback whale vocalizations was modified to detect blue whale D calls. The detector was set to search for frequency-modulated signals in the 35-90 Hz range where D calls typically occur (Thompson et al. 1996; Oleson et al 2007). There was a high false positive rate and subsequently a trained analyst used a detector evaluation tool titled GPLReview (github.com/MarineBioAcousticsRC/GPLReview) to reject false detections. Detections yielded start times, start frequency, end frequency and slope for all D calls.

iii. D call sequence analysis

A D call sequence was defined using initial observations of sequences on July 22nd 2019 where 28 sequences were observed (Figure 2). Using the distribution of inter-call intervals within a sequence and number of calls per sequence, D call sequences were filtered out of acoustic data from July 24th 2007 to October 29th 2020 for all data available at site H. D call detections with an inter-call interval between 5 to 10 s were considered potential sequences. Only those with more

than 4 calls were retained in the final analysis. The number of sequences per day and per year were summarized.

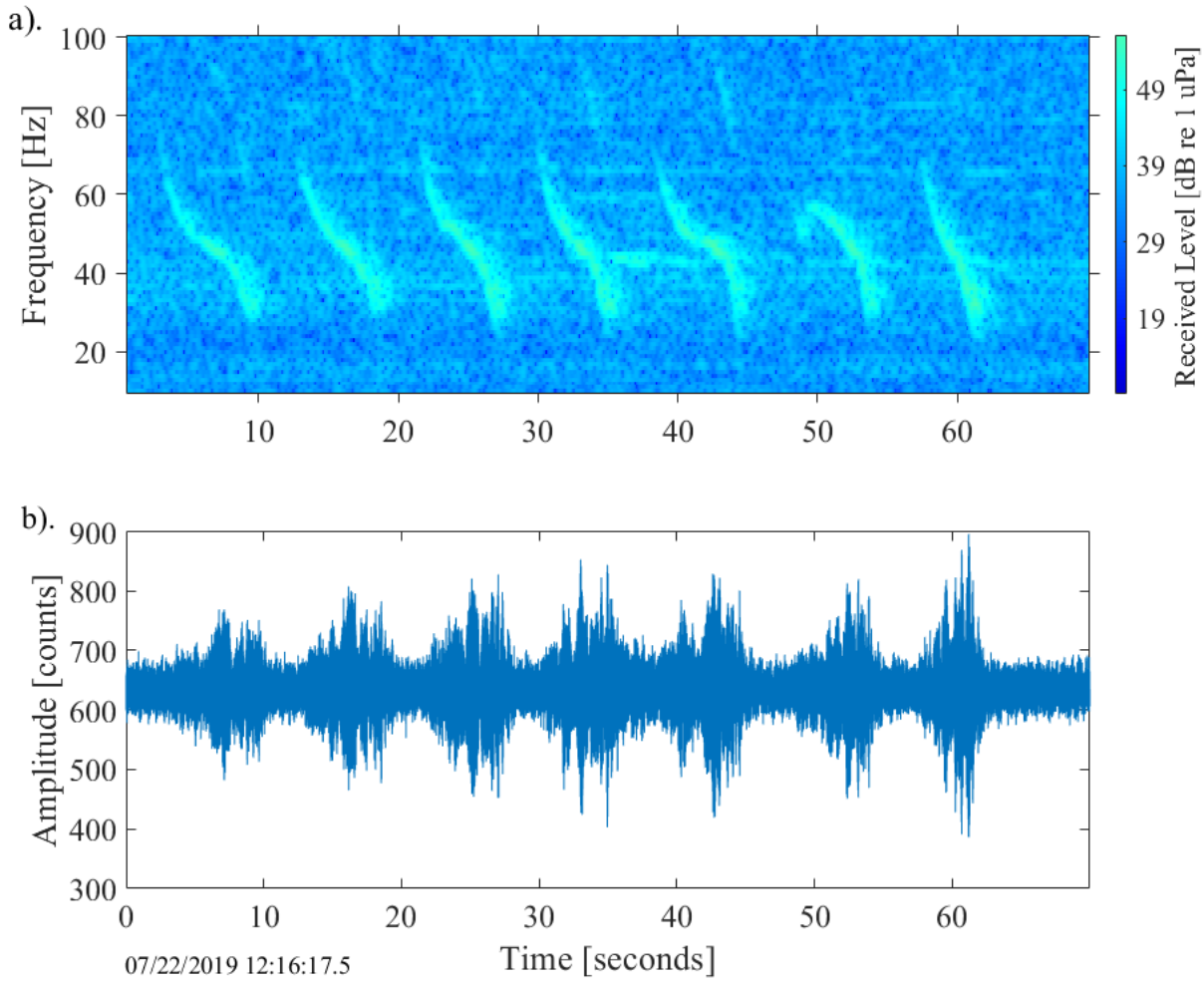

Figure 2: D call sequence example (a) spectrogram and (b) time series. One D call sequence on July 22nd 2019 at 12:16:17 UTC. Color in spectrogram represents received level after transfer function was added.

iv. Cross correlation for Time Difference of Arrival (TDOA) calculations

Using the start of a call arriving at each instrument, it is feasible to compute a time difference of arrival between pairs and estimate a location in 2D space. First, we needed to synchronize the clocks on all four instruments. For HE and HW, the systems were recording

upon recovery, hence it was possible to determine the time offset between the start and end of the deployment and calculate clock drift over the deployment period assuming a linear drift.

The HARPs at HS and HN had stopped recording upon recovery and a drift rate was calculated using the transponder pings observed on the acoustic data emitted during the HE (on 06/01/2019) and HW (on 07/13/2019) localizations. Using the HE localization, drift rate was calculated for both HS and HN. Expected arrival times were computed using a ping produced by the transponder on HE and the slant range from the HE transponder to the hydrophone on HS/HN. Sound speed at depth was calculated using a MicroCat at 20 m above the seafloor which collected temperature and conductivity data. This sound speed, 1485.1 m/s, was then used to calculate the time it takes for the transponder ping to travel directly from the source to the hydrophone at HS/HN. By subtracting the expected arrival time and the actual arrival time of 20 pings, a drift rate was computed for HS and HN. The HW localization was used to calculate a second drift rate for HN to verify this method. An HS drift rate could not be calculated from the HW transponder pings because there was no direct path from the HW transponder to the HS hydrophone. A second drift rate was calculated for HN using the same method as the HE localization and then drift rates were averaged to get the final drift rate for HN. The difference in drift rates for HN from both localizations was only 6.9e-09 s/s.

For the D calls, a 2nd-order Butterworth bandpass filter was applied to all the data on July 22nd 2019, using corner frequencies of 30 and 80 Hz. Using the start time from the D call detections, five seconds of data after the start time were extracted from the filtered data. Calls that were not identifiable on all four instruments were not further evaluated. Cross correlation was performed on the first second of each call by overlapping the time series and matching the waves in each hydrophone pair. The time series was manually adjusted to make sure only the

first second of each call was used in the cross correlation. These cross-correlations identified a time lag corresponding to the correlation maxima and hence a time difference of arrival for each of the hydrophone pairs. With an array of four instruments, 6 TDOAs were calculated for each unique pair of instruments (HE-HW, HE-HS, HW-HS, HE-HN, HW-HN, HS-HN). Only using the first second of each call allowed for more precise estimation of the time difference of arrival by limiting the amount of constructive and destructive patterns interfering with decisions on each cross-correlation maximum.

iv. TDOAs to location

In July 2017 and 2018 an instrument package that measured conductivity, temperature, and depth (CTD) profiled water properties at site H. The temperature and salinity data were averaged using both years and then averaged over the first 100 m and sound speed was calculated using the Chen-Millero equation (Chen and Millero 1977), resulting in a measured sound speed of 1485 m/s. The largest distance in the large aperture array was between sites HS and HN (1485.7 m), hence the maximum possible time traveled between instruments was 0.99 s. Any TDOAs calculated from the cross correlation that exceeded +/- 0.99 s were considered erroneous and excluded from further analysis.

The method used to compute locations for each call was the least-square best-fit grid search method (e.g., Wiggins et al. 2013). Expected TDOAs with a 10 m resolution were calculated for a maximum grid size of 5000 m x 5000 m using the sound speed, the locations of the instruments and a source depth of 30 m based on previous studies of calling depth (Oleson et al 2007b). The expected TDOAs were then compared with the calculated TDOAs from the cross

correlation and the location with the smallest difference between expected and calculated was chosen as the location of the call.

v. Source level calculation

Source level was calculated for all D calls that were localized using the equation:

$$SL = RL+TL \qquad \text{(Equation 1)},$$

where SL stands for source level, RL for received level and TL for transmission loss (Urick 1983). Only sites HE, HW and HS were used in the source level calculations due to disk write noise at site HN that would affect received levels. Source level was calculated for each site and then averaged for each call to get the final source level values. The received level was calculated from the raw data using an elliptical filter from 10-100 Hz to make sure that energy from higher frequency sources was not included. However, to make sure only the call was used to calculate received level, the energy of the time window was integrated and only the time in which 5% to 95% of the energy occurred was used in the calculation, similar to Keen et al 2018. For D calls, an 8 s time window was used to make sure the entire call was included in the integration of energy. The time window was extracted by subtracting 2 seconds from the D call start time and getting 8 seconds of filtered data from that time. To remove any edge effects, the first 2 seconds of the time window for D calls were removed, meaning 6s of data was used in the integration analysis. Received level was calculated in three different ways: peak to peak, zero to peak and root mean square. Counts were converted to a decibel value using a transfer function adjustment specific to each of the four hydrophones. Since the transfer function contained a specific value for each frequency, the average value over the 30 Hz to 80 Hz range was used.

Transmission loss includes the loss due to spreading and attenuation over distance. At low frequencies (less than 100 Hz), sound can travel far without attenuation having much of an effect. Hence, transmission loss was calculated using slant rage, assuming spherical spreading to estimate transmission loss (Keen et al. 2018). Slant range was calculated using the location of each call and the distance between the presumed source at 30 m depth and the hydrophone depth for each instrument. This number was then converted to decibels using the equation:

$$TL = 20\log_{10}(\text{slant range [m]}) \qquad \text{(Equation 2).}$$

vi. Error analysis

Error was calculated in two different ways: (1) error from the least-squares best fit TDOA to location model and (2) by looking at the physical limitations of the model. First, the model error was calculated as the square root of the squared difference between the expected and calculated TDOAs (1). Final locations included calls that had an average error less than 0.22s (Figure 3a). This error can then be converted to a value in meters using the sound speed. The second error analysis was conducted by adding noise to TDOAs with expected locations. Error in meters was then calculated as the difference between the expected location and the calculated location from the perturbed TDOAs (2). We defined noise as adding 22 ms (1/45Hz) to each TDOA in both the positive and negative direction, leading to a total amount of twelve noise perturbations. This number was derived from the peak frequency of D calls being 45 Hz. Therefore, the added noise simulated the cross correlation with one wavelength misaligned. The locations of the instruments as well as sound speed averaged over the first 100 m (1494 m/s) were also input into the simulation to document this specific array's performance. This resulted in an error map in which, using a ten-meter resolution, we can verify the location reliability in respect to their position to the array (Figure 3b).

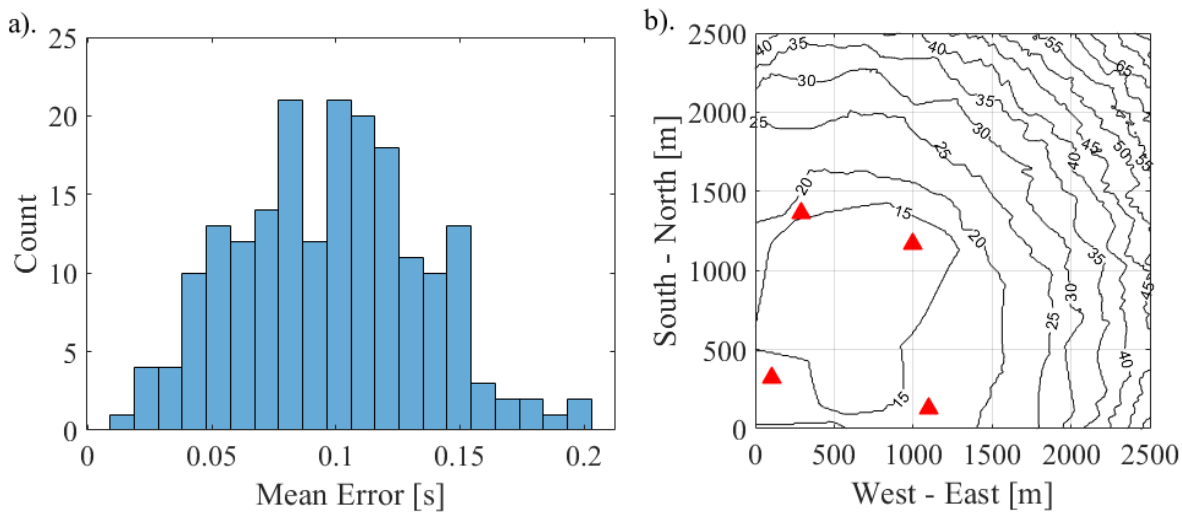

Figure 3: (a) Location error calculated from the model (1) and (b) error map based on physical limitations (2). (a) Histogram showing error in seconds for 194 D call localizations **(b)** Error map calculated using 22 ms noise perturbations and 10 m resolution for a 2500 x 2500 grid with sites in red.

ANALYSIS

i. D call sequence analysis (2007-2020)

D call detections at Site H from 2007 to 2020 were analyzed using the GPL detector and manual verification by analysis. There was a total of 77,174 D call detections from July 24th 2007 to October 29th 2020 at the study site (Figure 4a). Counts per day and counts per year were computed using all D detections. There was a max number of all D call detections on July 29th 2019 (1,746 calls). The max number of D calls per year occurred in 2017 with 15,114 calls with a minimum of 449 calls that occurred in 2013 (Figure 5a).

All Site H detections were filtered to only include calls with inter-call intervals greater than 5s and less than 10s and then only sequences greater than 4 were included in the sequence analysis for this data. This was done to see if the D call sequences appear in other years in the data. Out of 77,174 total D call detections (Figure 4a), 963 of them are D calls a part of a sequence with 165 total sequences (Figure 4b). The average inter-call interval of calls within the

sequences is 7.29s +/- 1.32s. The average number of calls per sequence from 2007 to 2020 is 5.8 calls +/- 1.1 call. There was a max number of sequences on July 22nd 2019 where 35 sequences occurred, 28 of which were the ones between 12:00 and 14:20 UTC (Figure 4b). The year 2019 had the greatest number of sequences with 63 sequences (Figure 5b). The minimum number of sequences per year was 1 sequence in 2009 and 2013 (Figure 5b).

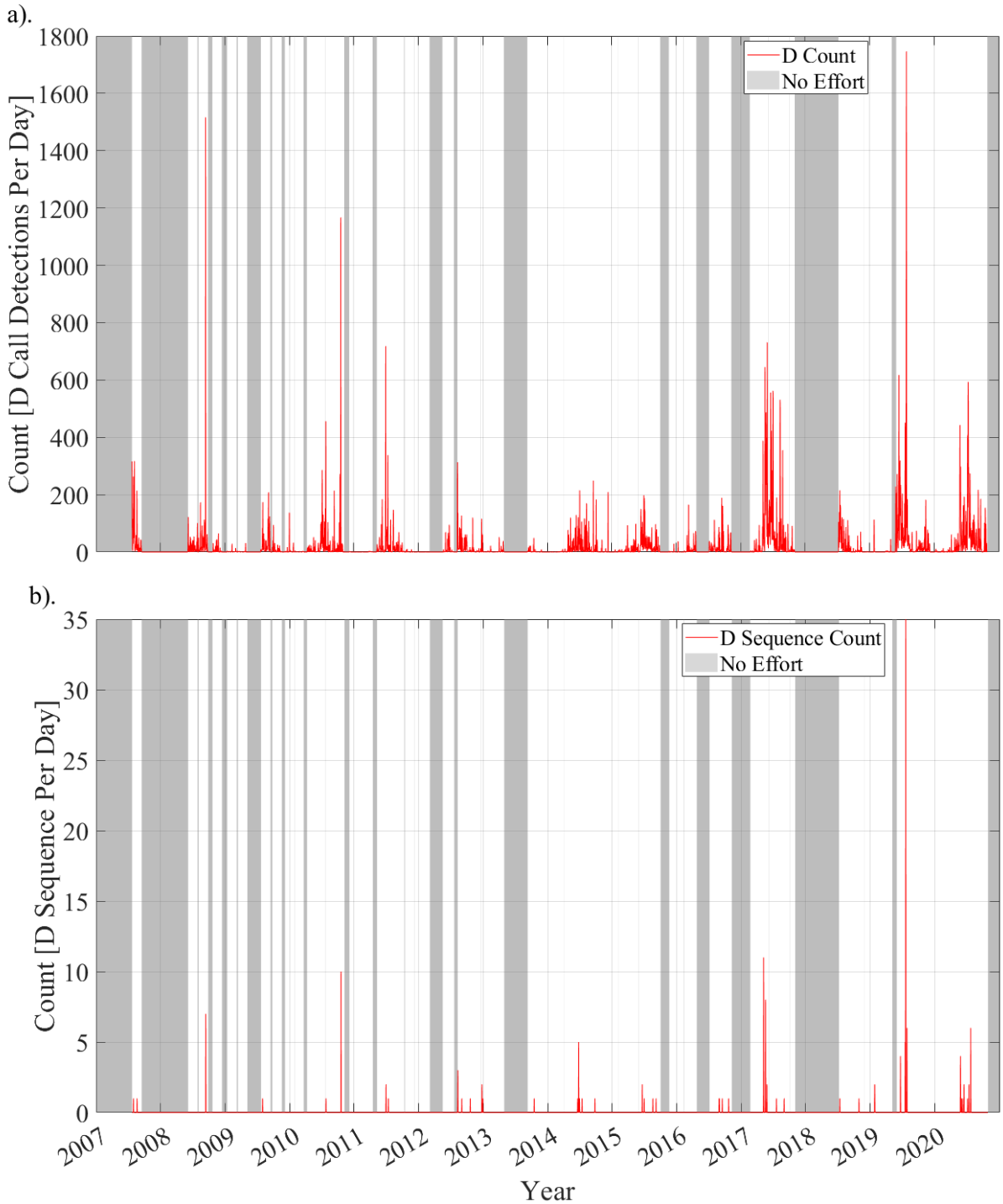

Figure 4: (a). All D detections count per day and (b). D sequence count per day. Count per day for (a) overall D call detections and (b) D sequences from 2007 to 2020. Grey shaded regions indicate times where there was no effort at the site.

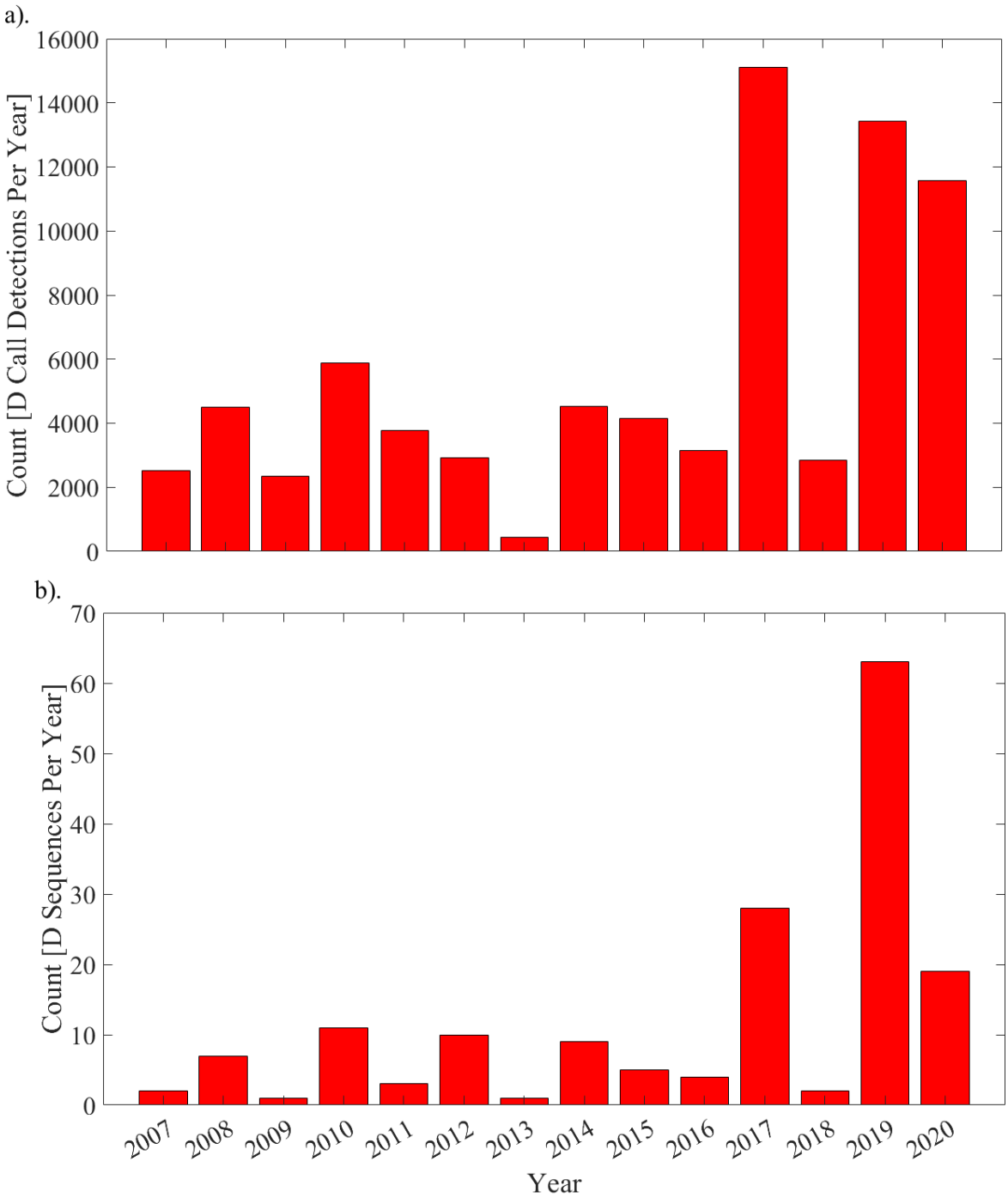

Figure 5: (a). All D detections count per year and (b). D sequence count per year. (a) All D detections per year from 2007 to 2020. (b) Count of D sequences per year from 2007 to 2020 using all D detections data.

ii. Localized Sequences on July 22nd 2019

Further analysis was performed on data from July 13th 2019 to July 27th 2019 where four instruments were recording as a large aperture array (Table 1, Figure 1b). The number of D calls per day revealed that July 22nd had more than twice the number of the calls than any of the other days (451 calls) (Figure 6). Further analysis of this date revealed 207 D calls from 12:00 h UTC to 14:20 h UTC (Table 2). Displaying the calls using spectrograms in Triton (Wiggins and Hildebrand 2007) revealed sequences of multiple D calls in a 60 s to 90 s period every 3 to 5 minutes. Each sequence of D calls had an average inter-call interval of 7.3 s +/- 1.7 s with on average 7.3 calls +/- 1.4 call per sequence. There was a total of 28 D call sequences in this period and sequences ranged from a minimum of 5 calls per sequence to a maximum of 10 calls per sequence. Sequences were also analyzed visually using spectrograms in Triton to verify that call reflections were not included in the sequence count. Each call in a sequence had a similar start and end frequency, duration, and slope, as well as a uniform inter-call interval with minimal variability (Figure 2). This initial observation of sequences was used to define parameters with which to look for further sequences in the larger dataset.

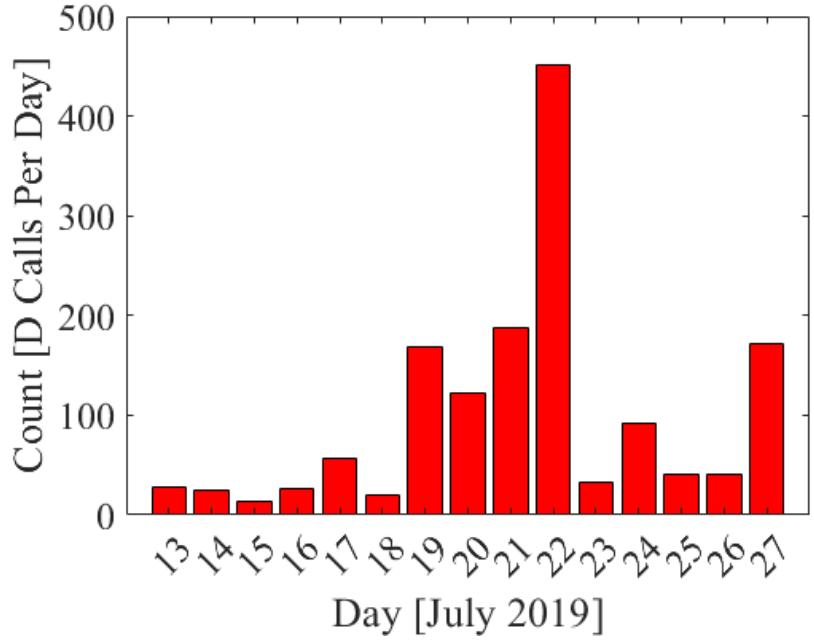

Figure 6: D call count July 13th to July 27th. Histogram of D call count per day for the overlapping recording period of the large aperture array.

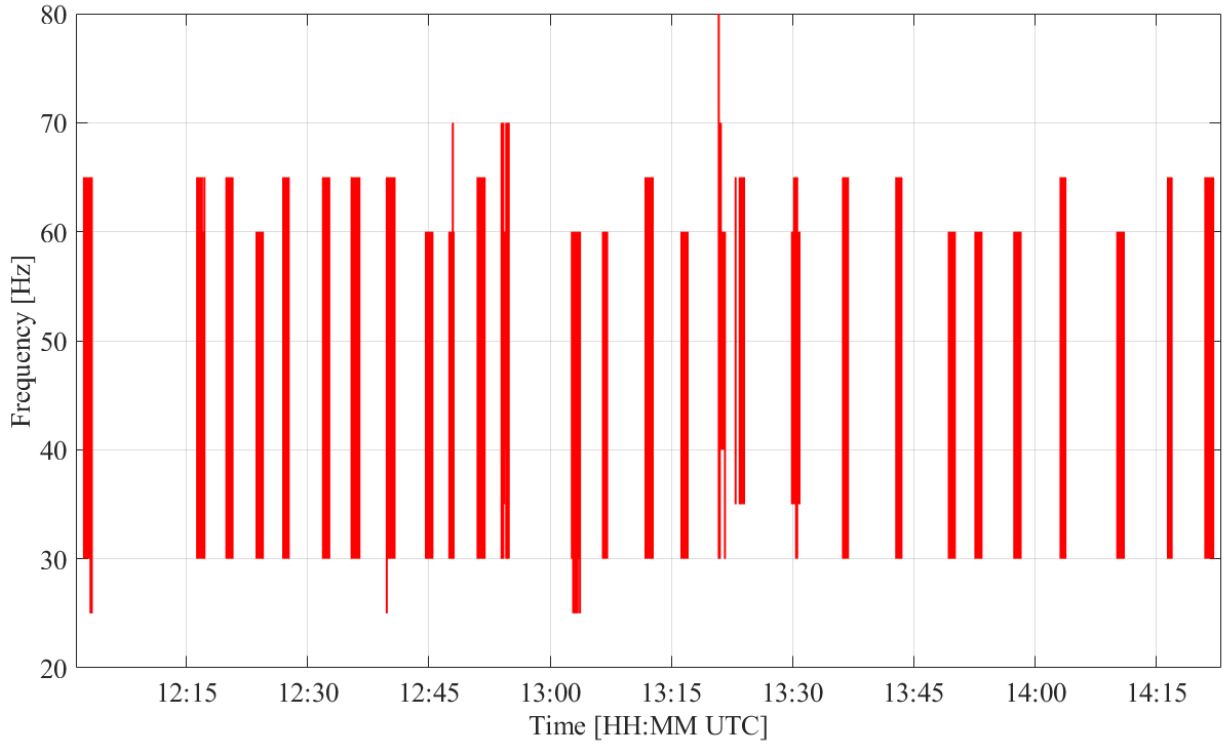

Figure 7: D call start times on July 22nd, 2019, 12:00 UTC – 14:20 UTC. Each individual line indicates the start time for D calls in red. Vertical length of the bars show the frequency range of each call.

Table 2: Call characteristics for D calls on July 22nd, 2019. Average +/- standard deviation start frequency, end frequency, duration, and inter-call interval for each D call (within and between sequences).

	Number of Calls	Start Freq (Hz)	End Freq (Hz)	Duration (s)	Inter-call Interval (min)
2019-07-22 12:00 UTC to 14:20 UTC	207	65 +/- 5	32 +/- 4	4 +/- 1	0.6 +/- 1.3

iii. Locations

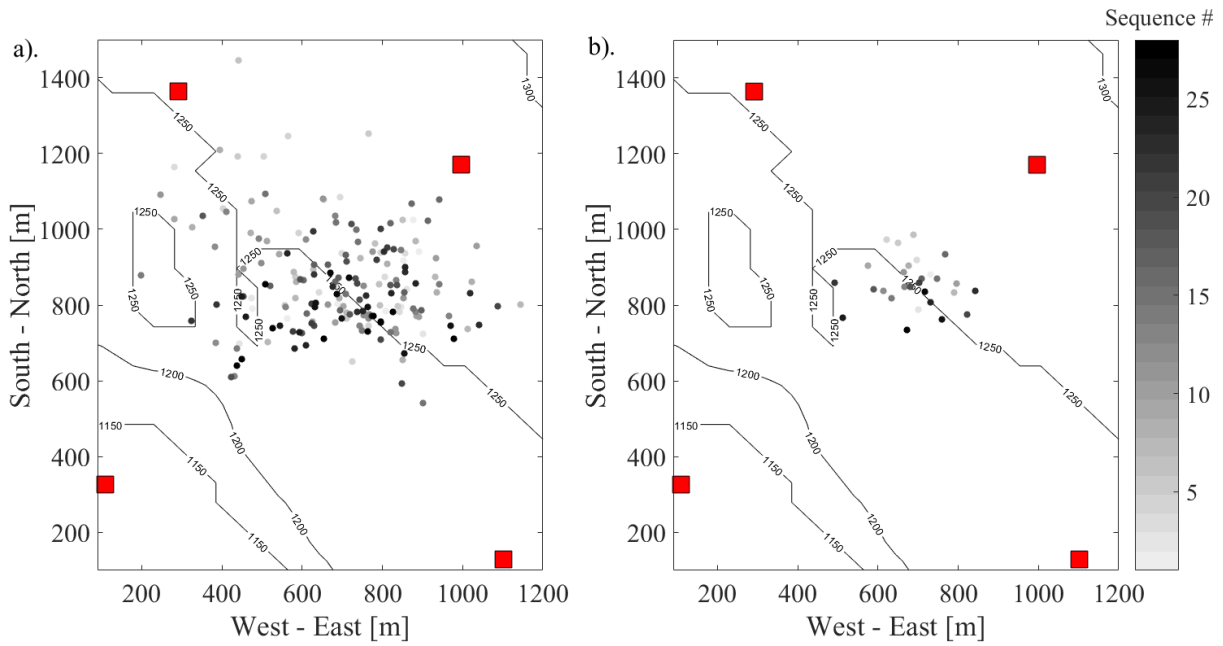

Figure 8: (a) Location map with 1 D call per location and (b) Location map with average location per sequence. Both plots have sites in red and locations colored by sequence number in which lighter greys indicate earlier sequences and darker greys indicate later sequences.

After filtering out erroneous calls and locations, 194 D calls were used in the localization on July 22nd 2019, 12:00 h to 14:20 h UTC (Figure 8a). The maximum error for the D call locations was 0.203 s with an average error of 0.098s +/- 0.037 s (Figure 3a). Locations were computed for all D calls from and average location per sequence was calculated to minimize the variability within each sequence (Figure 8b). All D call locations are found inside the array except for three D call locations outside the array (Figure 8a). The average per sequence

locations were all found inside the array in a 250 m by 350 m area centered on the 1250 m depth contour (Figure 8b).

Further analysis was performed on three sequences to understand individual sequence cohesion spatially (Figure 9a). Since each call within a sequence was only ~7 seconds apart, if each sequence was an individual whale, we would expect the locations to be close in space. Assuming the speed of a blue whale while calling to be the max of the range of 1.3 to 2.5 m/s (McDonald et al 2001), the blue whale should only travel ~17.5 m to its next call location within a sequence. Error was calculated for each call within the three sequences in which a circle of error in meters, including both model error and physical error, was plotted around each location. An example of this is shown in Figure 9b, in which one sequence at the end of the period was plotted with error circles. One of the locations was shown to be 83 m apart from the next location, which is much more than expected. However, the overlapping error circles display that the locations may be closer than they appear (Figure 9b). Out of the 28 sequences, 9 of them had one D call at the start of the sequence with a location that did not have an overlapping error circle with the other locations. These locations were removed from the rest of the sequence analysis. Each sequence had locations in a similar area, but locations were not as close as expected which is likely due to the location error (Figure 3).

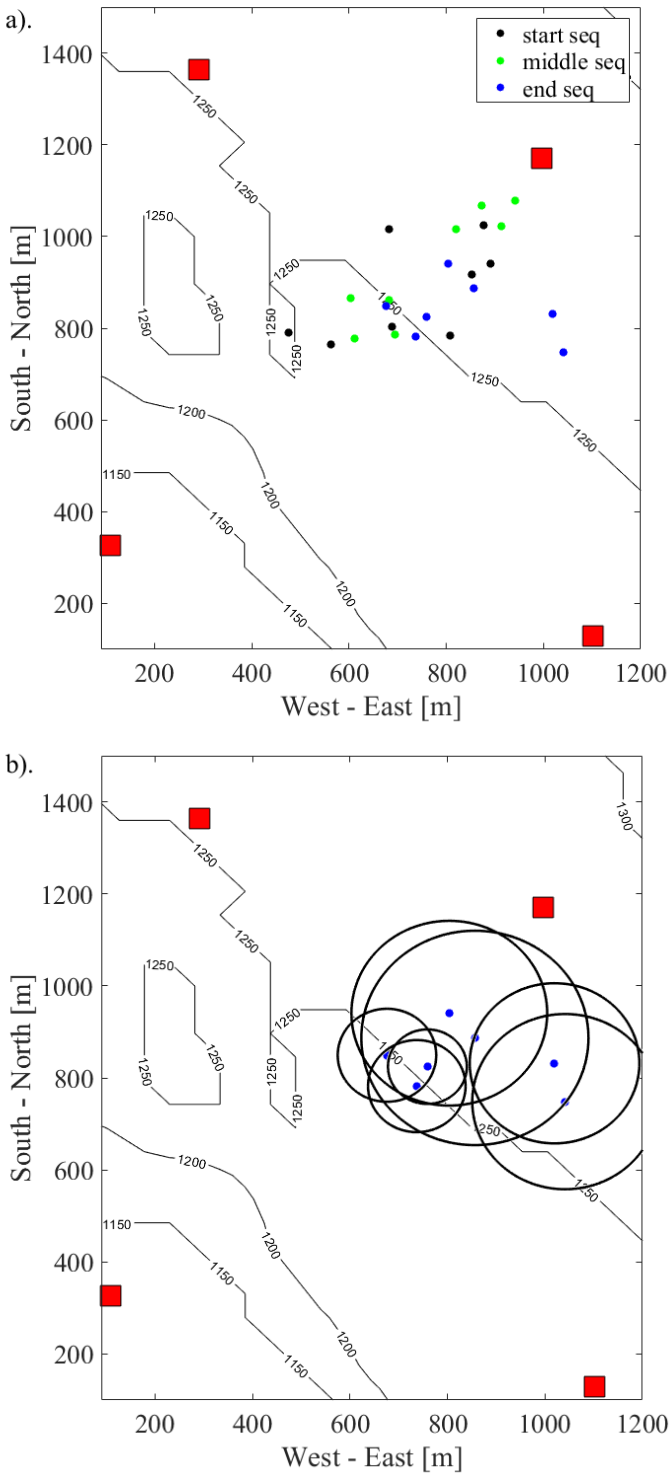

Figure 9: (a) Location map with 3 sequences and (b) Location map with end sequence and error plotted. Sites are shown in red for both plots. (a) Locations for 3 sequences at the start, middle and end of the period localized. (b) End sequence only with circles with an error radius calculated by combining both error analysis methods (Figure 3).

iv. Source levels

Table 3: D call received and source levels. Received levels (RL) and source levels (SL) using peak to peak (pk-pk), zero to peak (0-pk) and rms calculations.

Call	RL type	RL (dB re 1uPa)	SL (dB re 1uPa at 1m)
D	pk-pk	114.5 +/- 1.6	177.3 +/- 1.9
	0-pk	109.9 +/- 1.6	172.7 +/- 1.8
	rms	99.5 +/- 1.7	162.3 +/- 2.2

For D calls, source level was calculated over the 30 Hz to 80 Hz frequency range using three different calculations. The rms received level values for all 194 D calls used in localization was highest at HW (101.4 +/- 2.6 dB re µPa) and lowest at HS (97.9 +/- 2.6 dB re µPa). Slant range was highest at HS (1479.9 m +/- 99.94 m) and lowest at HE (1315.9 m +/- 51.1 m). A linear fit was applied to the rms received level data in relation to the logarithmic slant range. This relationship resulted in a negative slope and an R^2 of 0.05 (Figure 10a). Average D call rms source level was 162.3 +/- 2.2 dB re 1 µPa at 1 m (Figure 10b, Table 3). The zero to peak calculation resulted in a source level of 10.4 dB re 1 µPa at 1 m higher than rms and peak to peak source level was 15.0 dB re 1 µPa at 1 m higher than rms (Figure 10b, Table 3).

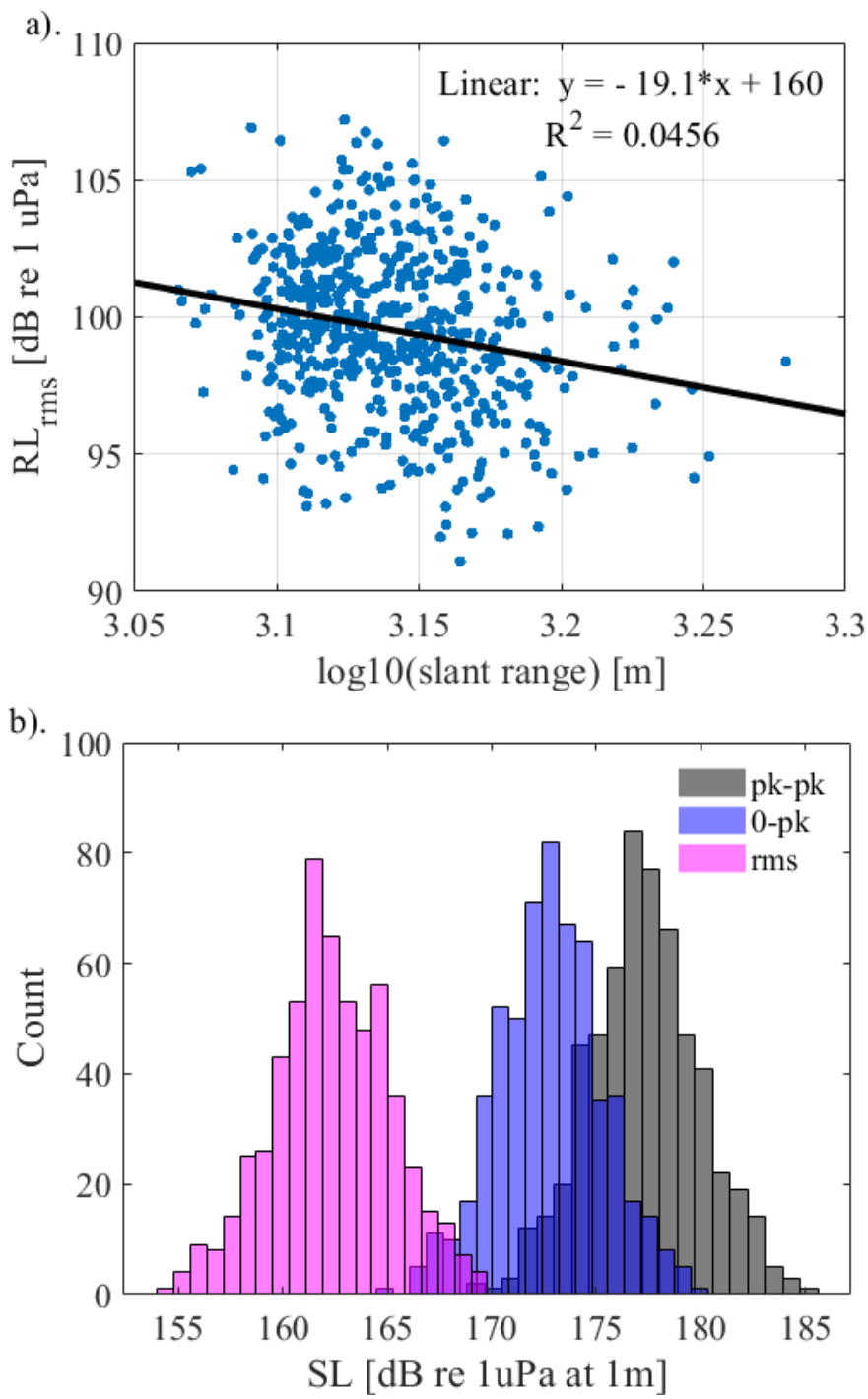

Figure 10: (a) Received level (rms) vs. slant range and (b) D call source levels using different received level methods. (a) Black line shows linear fit for all the rms received level data when all sites are combined with equation and r-squared value displayed on the plot. (b) Source level histogram for each received level type shown in different colors.

DISCUSSION

i. D call sequences

D call sequences described in this study have not been documented as a part of the blue whale sound repertoire before. A and B calls produced by blue whales have been shown to be produced in sequences with 40 to 50 s intervals to form song (Oleson et al. 2007). Fin whales produce 20 Hz calls in regular sequences with 15-25s in between calls (Weirathmueller et al. 2017; Širović et al. 2017). Both fin whale 20 Hz song and blue whale A/B song have been previously associated with reproduction to attract mates (Širović et al. 2017; McDonald et al. 2006). Both these baleen whale species also produce a separate call type associated with foraging (fin whale 40 Hz and blue whale D calls), but no song patterns relating to foraging for either whale have been documented (Romagosa et al. 2021; Wiggins et al. 2005). It is possible that the D sequences have a specific function that has not yet been documented.

There have been short sequences of D calls mentioned in previous studies. However, these studies refer to multiple whales calling at the same time where calls at different frequencies are seen 5-10 s after each other (Oleson et al 2007; Mcdonald et al. 2001). These studies indicate that D calls can be used as contact calls and blue whales may have individual-specific D call characteristics. D calls can be produced in overlapping pairs each produced by a different animal commonly referred to as call-counter-call (McDonald et al. 2001). The overlapping period and different frequency content of pairs of D calls related them to two different individuals. All the sequences had similar start and end frequencies and similarly shaped D calls, indicating that one blue whale may be producing each of the sequences. The consistent inter-call interval further indicates that it is likely an individual whale produces each sequence. This is further shown in Figure 8b where the average locations of each sequence over a two hour and a half period are in

a relatively small area (250 m by 350 m). It is possible that an individual whale is producing all the sequences, but this hypothesis may require other methodologies to prove.

In the long-term acoustic data at the site, D call sequences are visible from 2007 to 2020. This shows that sequences have previously been overlooked and have been in the blue whale repertoire for at least those 14 years. However, out of the 77,174 total D call detections only 1.2% were part of a sequence. It is possible that blue whales only use sequences when performing a specific behavior or only a small group of the population produces these sequences. The number of sequences does seem to increase throughout the years with a higher number of sequences in 2017 (17.0%) and 2019 (38.2%). Those two years also had the highest number of total D detections with 19.6% occurring in 2017 and 17.4% occurring in 2019. Although there were a greater number of total D calls in 2017, 2019 had the greatest number of sequences. This may be because D sequences have a different function than singular D calls, however more investigation needs to be done to confirm this. More investigation into the characteristics of the sequences throughout the 14 years may help in determining the possible behavior related to the sequences.

The relatively scattered locations of the D calls were what we expected for a call previously associated with foraging and social interaction (Oleson et al 2007). Since D calls have been shown to be produced by whales traveling in pairs and in groups in between foraging dives, it is possible that there was a good prey patch within our array that caused the whale(s) to linger at the site (Calambokidis et al 2008). Concrete measurements of the environmental conditions and prey availability at the site would help support such a hypothesis. Previous research has shown that blue whales can stay up to 34 hours in a particularly abundant prey patch, so the whales being at the same location for more than 2 hours is conceivable (Irvine et al

2019). This knowledge of a consistent foraging ground would be important in understanding how we can protect these animals from human activities such as ship transits.

ii. Location assessment

The error map from the error analysis shows that there was a higher error in locations outside of the array (Figure 3b). There was lower error closest to the east instrument than any of the three instruments which likely is due to the deeper depth of this instrument and the sound traveling downslope rather than up. This means that it was more likely that sound traveled in a direct path towards HE which minimizes error in the slant range calculation. All D call locations in this study were found to be inside the array which is suspicious because it is unlikely that the whales specifically happened to be found only inside the array. However, since locations with higher error were filtered out, locations that were outside the array may have been filtered out in the process. This could also be explained by the method of picking the calls used in localizations because only calls with high received level were used as they tended to have a better signal to noise ratio. The method used in Miller et al. 2021 showed similar results, in which on different days D call locations were clustered in small areas (Miller et al. 2021). This was the only other study that used D calls for acoustic localization, however the similar patterns in both that study and this one is promising.

Locations were further assessed in two different ways: using the D call sequences and received level of the calls. Since the D call sequences have D calls that have the same frequency content, shape, and regular ~7 s intervals, it is highly probable that each sequence is produced by an individual whale. Therefore, the locations for each sequence should be relatively close together in space, given the swimming speed of 1.3-2.5 m/s for calling blue whales (McDonald

et al 2001). An individual blue whale producing a sequence should not move more than 17.5 m to the next D call location. Error circles using both the calculated model from the error and the physical error from the error map were calculated and locations were verified taking error into account. Only 9 out of the 28 sequences had a location that was far away from the other locations in the sequence. Further analysis into the sequences themselves would help verify if those erroneous locations were produced by a single whale or were from another whale within the error range. Overall, the sequence locations were used to make sure the method was working based on how close most of the sequences were in space.

Received level was also used to verify locations. Received level was plotted against range similar to Figure 6 in Wiggins et al. 2021 (Figure 10a). If the locations were accurate, the received level should decrease with range due to sound being more intense as the animal gets closer to the instrument. When the peak to peak received level for D calls were plotted against logarithmic slant range, a linear line was fitted to the data to see if there was a negative slope. The D call data had a slope of -19.1 m with an R^2 of 0.05. Therefore, the received level does decrease with slant range although the low R^2 value shows there was a lot of variability within the data. Previous studies documented that signals decrease in received level at a slower rate at shorter ranges which may explain the small slope in our data (Wiggins et al 2021). Since most of the locations were close to the instruments there was a larger number of calls at a similar range which may be affecting the linear fit. Overall, the unique patterns in this data allowed for multiple assessments of the locations of the data.

More investigation needs to be done into error caused by the cross correlation, clock sync, and inputs put into the expected TDOA model used in the least-square best-fit grid search method. The cross-correlation method was used mainly because the start time of the D calls were

difficult to pick manually due to the difference in energy on each of the four instruments and destructive and constructive interference notable in the call timeseries. Developing a technique to manually pick the D call, for example picking the start at the highest energy rather than when the signal first arrives, might help reduce some of the cross-correlation error. Using the knowledge derived from the results of this study can help us perfect the method for D call localization in the future.

iii. Source level

Previous manuscripts on blue whale source levels have mainly used blue whale B calls to calculate source level (Samaran et al. 2010; Bouffaut et al. 2021; Sirovic et al. 2007; Thode et al. 2000). In this study, we calculated source level using three different approaches. In previous studies, rms pressure was used to compute blue whale source level (Bouffaut et al. 2021). However, rms pressure is highly dependent on the time window used to compute the value and therefore may become difficult for comparison. This is the most used approach for tonal signal level computations. Alternative calculations may compute peak-to-peak or zero-to-peak source levels, which are more often applied to impulsive signals. For the same waveform, decibel levels may vary by 10 dB or more depending on the received level metric of choice (Madsen 2005). This can be seen in this study, in which for D calls the source level average for peak-to-peak was 15.0 dB higher than the average for rms source level. Both the distributions for the peak to peak and zero to peak source levels were higher than the entire rms source level distribution (Figure 10b). This is due to the complexity of the D calls caused by multipathing. The peak to peak and zero to peak received levels were computed from instances where constructive interference occurred in the D calls, causing the levels to be higher than they would be without interference.

Therefore, rms source level is the best estimate for D calls as complex as the ones in this study. The method used to calculate transmission loss is also important to note as sound does not travel in the same way in different regions due to different waters having different sound speed profiles. In this study, all locations had a slant range less than 2000 m, over which the signal was able to travel in a direct path, meaning the best method was to assume spherical spreading loss (Equation 2).

A study focusing on blue whale and fin whale source level in the Antarctic included a table of source levels from previous publications, but also included D call source level estimations (Table 1, Miller et al 2021). This paper estimated D call source levels in the Eastern North Pacific by assuming a 40 Hz bandwidth and adding 16 dB to levels reported in Thode et al. 2000. The D calls reported include a wide range of values with the lowest being 142 dB re 1 µPa at 1 m rms (North Atlantic) and the highest being 190 dB re 1 µPa at 1 m rms (Antarctic). The source levels reported for D calls in this paper were within the range of previously reported D call source levels. The average D call rms source level calculated over the 30-80 Hz frequency band was 162.3 +/- 2.2 dB re 1 µPa at 1m. This number was about 14 dB lower than the 176 dB re 1 µPa at 1 m rms minimum reported for the Eastern North Pacific region (Miller et al 2021). Since the source levels in this study were computed using localized D calls with bottom moored hydrophones, to our knowledge, this study presents the most reliable rms source levels for the Eastern North Pacific blue whale population. Further improvement of D call localization and standardization of the method used to calculate rms received level can help us refine these source level calculations in the future.

CONCLUSION

This study documents novel D call sequences. D calls as part of a sequence had an inter-call interval of 7.3 s +/- 1.7 s with on average 7.3 calls +/- 1.4 call per sequence. D call sequences were identified in all years of data analyzed and hence have been present in the blue whale sound repertoire at least since 2007. Further investigation may shed light on potential annual, seasonal, or daily trends regarding when sequences are produced preferentially. The presence of the D call sequences highlights the importance of not only looking at large time periods of data but short periods of time as well. Out of all the D call detections, not many may be a part of a sequence, but those that do, show a unique pattern that may help in the identification of an individual whale or behavior. Understanding call behavior and how often blue whales produce calls is important for density estimation analyses and the presence of this sequence adds another complication to call count estimates.

The D call sequences localized on June 22nd 2019 produced locations within the array indicating that there may be a possible abundant prey patch in the area. Assuming that the sequences were produced by a single individual, locations were too far apart given their average inter-call interval and possible distance traveled. The complexity of the D calls caused by interference patterns may be the cause of this error. Estimating interference patterns using sound propagation models of the environment may be a better way to inform us on the call's location and can help in improving this process in the future.

The findings in this thesis highlight different methods for studying blue whales using passive acoustic monitoring. By combining both observations of calling behavior and localizations, we can gather insight into the temporal and spatial behavior of blue whales when emitting D calls. This approach can be applied in future efforts to acoustic data from other locations or years to investigate variability of time and space. Source levels reported in this study

can be used for density estimation, localization of blue whales using propagation methods, or noise impact studies. Understanding blue whale calling behavior, with the use of long-term passive acoustic monitoring, can help us protect these large animals without interfering with their natural environment.

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